

Loyola Marymount University

CMSI 3300 Artificial Intelligence

Fall 2024

Professor Lanyu Shang

Announcements

- Homework 1
  - Due **this Friday** 10/4 @5 pm
- Classwork 4
  - Available on GitHub Classroom/Brightspace
  - Due next Friday 10/11 @5 pm
- Midterm Question Contribution
  - Google Form: https://forms.gle/PQWnzJ3EphU2WyPw8
  - Due next Thursday 10/10 @5 pm

This Lecture

- Value of Information
- Homework 1

Last Lecture

- Utility Theory: Probability -> Decision
  - **Utility Function** $U(s, a) = u$
    - mapping “state” and “action” to a “utility score”
  - **Expected Utility** $EU(a|e) = \sum_{s \in S} P(s|a, e) \cdot U(s, a)$
  - **Maximum Expected Utility**
    - $MEU(e) = \max_{a \in A} EU(a|e)$
    - $a^* = \arg\max_{a \in A} EU(a|e)$

Value of Perfect Information

- The **Value of Perfect Information (VPI)** computes the **expected increase in utility** from **acquiring new evidence.**
- Examples
  - BlindBot: whether to take an exploratory move
  - Monopoly: whether to buy a property

Computing VPI

- Input
  - Decision Network
  - Information/Evidence
    - Known evidence $E = e$ that is already available to us
    - Potential evidence E' that we do not yet have and are evaluating
- What is the Output?
  - **VPI**: difference in MEU between having and not having the new information E'

$$VPI(E'|E = e) = MEU(E = e, E') - MEU(E = e)$$

Computing VPI

MEU of acting optimally knowing E (what is available now) **and** E'

MEU of acting optimally knowing E (what is available now)

$$VPI(E'|E = e) = MEU(E = e, E') - MEU(E = e)$$





Goal: The **value of learning** about the state of E' if already know E

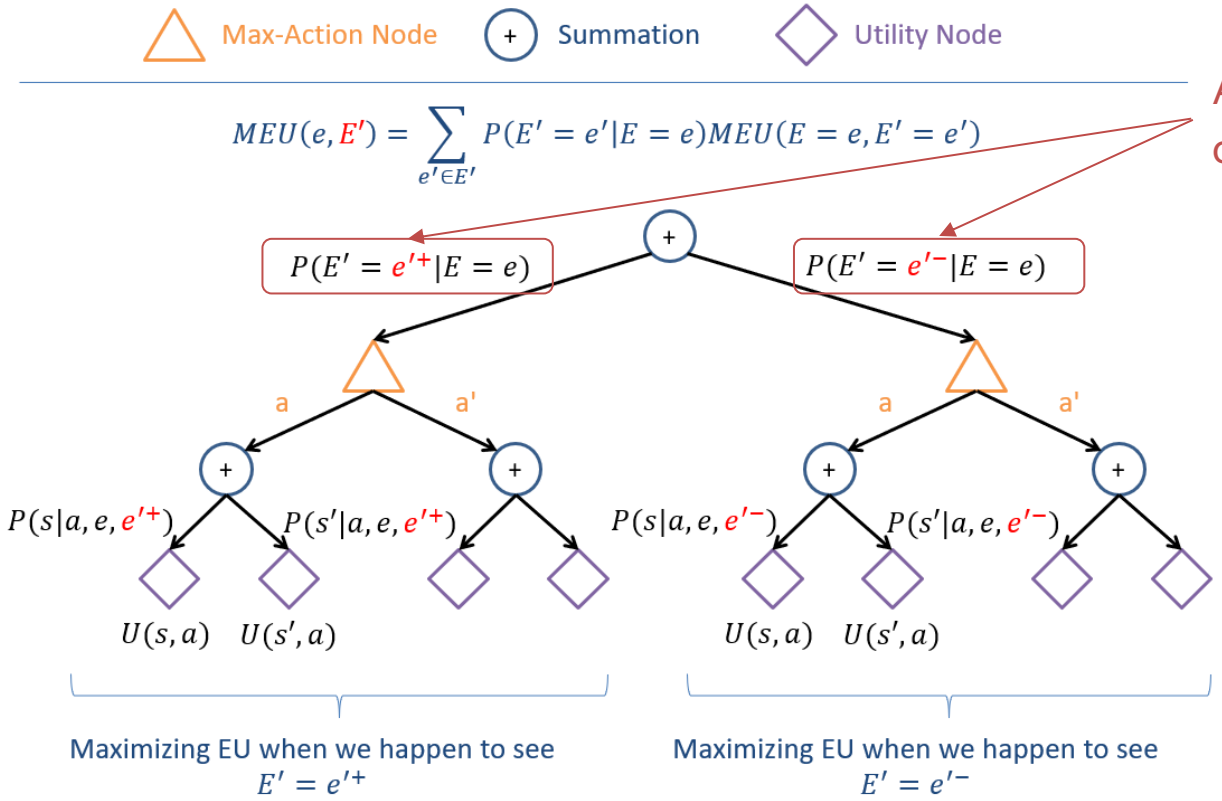
Idea: the weighted sum of MEU on all possible values of E'

Compute $MEU(E = e, E')$

- The Maximum Expected Utility associated with the reveal of some variable E' given some other evidence $E{=}e$ is the **weighted sum** over all values $e' \in E'$ of the likelihood of seeing e' with the MEU of how we would act seeing e', or formally:

$$MEU(E = e, E')$$
$$= \sum_{e' \in E'} P(E' = e' | E = e) \cdot MEU(E = e, E' = e')$$

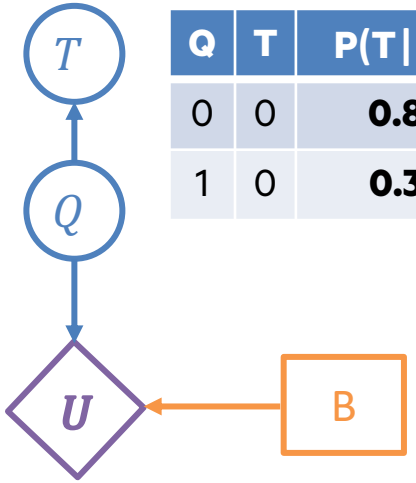
Visualizing VPI



Computing VPI

- **Example: used car test**
- The **quality** Q of the car can be in good shape (1) or bad shape (0).
- You may carry out a **test** T before you buy the car. The result of the the test is either pass or fail.
- A test will cost the buyer $200.
- **Task:** decide whether to take the test.

Q	P(Q)
0	**0.4**



Q	T	P(T|Q)
0	0	**0.8**
1	0	**0.3**

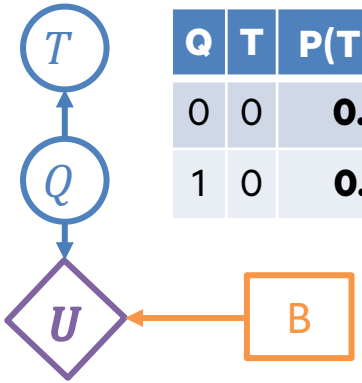
Q	B	U(Q,B)
0	0	**60**
0	1	**40**
1	0	**20**
1	1	**100**

Step 1: Compute $MEU(E = e)$

$$VPI(E'|E = e) = MEU(E = e, E') - MEU(E = e)$$

- **Step 1.0: Identify E**
  - No evidence: $E = \emptyset$
- **Step 1.1: Find $P(S|A, e)$**
  - $P(S|A, e) = P(Q|B) = P(Q)$
- **Step 1.2: Find $EU(a|e) \forall a \in A$**

$$EU(a|e) = \sum_{s \in S} P(s|a, e) \cdot U(s, a)$$



Q	T	P(T|Q)
0	0	**0.8**
1	0	**0.3**

Q	B	U(Q,B)
0	0	**60**
0	1	**40**
1	0	**20**
1	1	**100**

Q	P(Q)
0	**0.4**

Step 1: Compute $MEU(E = e)$

$$VPI(E'|E = e) = MEU(E = e, E') - MEU(E = e)$$

- **Step 1.2: Find $EU(a|e) \forall a \in A$**

Step 1.1: $P(S|A, e) = P(Q)$

$$EU(a|e) = \sum_{s \in S} P(s|a, e) \cdot U(s, a)$$

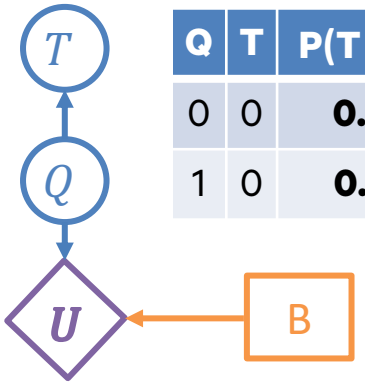
$$= \sum_{q \in Q} P(Q = q) \cdot U(Q = q, B = b)$$

$$EU(B = 0) = P(Q = 0)U(Q = 0, B = 0) + P(Q = 1)U(Q = 1, B = 0)$$

$$= 0.4 * 60 + 0.6 * 20 = \mathbf{36}$$

$$EU(B = 1) = P(Q = 0)U(Q = 0, B = 1) + P(Q = 1)U(Q = 1, B = 1)$$

$$= 0.4 * 40 + 0.6 * 100 = \mathbf{76}$$



Q	T	P(T|Q)
0	0	**0.8**
1	0	**0.3**

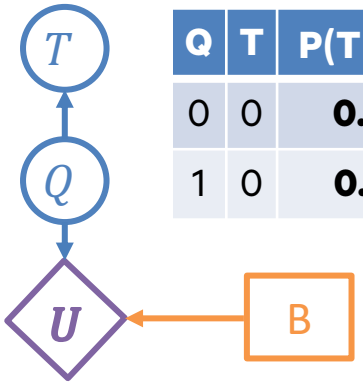
Q	B	U(Q,B)
0	0	**60**
0	1	**40**
1	0	**20**
1	1	**100**

Q	P(Q)
0	**0.4**

Step 1: Compute $MEU(E = e)$

$$VPI(E'|E = e) = MEU(E = e, E') - MEU(E = e)$$

- **Step 1.3: Find $MEU(e)$**
  - Step 1.2:
    - $EU(B = 0) = 36$
    - $EU(B = 1) = 76$
  - $MEU(e) = \max_a EU(a|e) = 76$



Q	T	P(T|Q)
0	0	**0.8**
1	0	**0.3**

Q	B	U(Q,B)
0	0	**60**
0	1	**40**
1	0	**20**
1	1	**100**

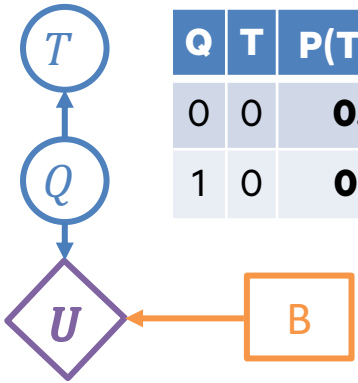
Q	P(Q)
0	**0.4**

Step 2: Compute $MEU(E = e, E')\forall e' \in E'$

- **Step 2.0: Identify E'**
  - $E = \emptyset$ and $E' = T$
  - $MEU(T)\forall t \in T$
- **Step 2.1: Find $P(S|A, e)$**
  - $P(S|A, e) = P(Q|B, T) = P(Q|T)$

Q	T	P(T|Q)	P(Q,T)	P(Q|T)
0	0	0.8	**0.32**	**0.32/(0.32+0.18)=0.64**
0	1	0.2	**0.08**	**0.16**
1	0	0.3	**0.18**	**0.36**
1	1	0.7	**0.42**	**0.84**

$$VPI(E'|E = e) = MEU(E = e, E') - MEU(E = e)$$



Q	T	P(T|Q)
0	0	**0.8**
1	0	**0.3**

Q	B	U(Q,B)
0	0	**60**
0	1	**40**
1	0	**20**
1	1	**100**

Q	P(Q)
0	**0.4**
1	**0.6**

Step 2: Compute $MEU(E = e, E')\forall e' \in E'$

- **Step 2.2: Find $EU(a|e)\forall a \in A$**

$$VPI(E'|E = e) = MEU(E = e, E') - MEU(E = e)$$

$$EU(a|e) = \sum_{s \in S} P(s|a, e) \cdot U(s, a)$$

$$EU(B|T = t) = \sum_{q \in Q} P(Q = q|T = t) \cdot U(Q = q, B = b)$$

Case 1: T=0

$$EU(B = 0|T = 0) = P(Q = 0|T = 0) \cdot U(Q = 0, B = 0) + P(Q = 1|T = 0) \cdot U(Q = 1, B = 0) = 0.64 * 60 + 0.36 * 20 = 45.6$$

$$EU(B = 1|T = 0) = P(Q = 0|T = 0) \cdot U(Q = 0, B = 1) + P(Q = 1|T = 0) \cdot U(Q = 1, B = 1) = 0.64 * 40 + 0.36 * 100 = 61.6$$

$$MEU(T = 0) = 61.6 \;\&\; a^* = \{B = 1\}$$

Q	T	P(T|Q)	P(Q|T)
0	0	0.8	**0.64**
0	1	0.2	**0.16**
1	0	0.3	**0.36**
1	1	0.7	**0.84**

Q	B	U(Q,B)
0	0	**60**
0	1	**40**
1	0	**20**
1	1	**100**

Step 2: Compute $MEU(E = e, E')\forall e' \in E'$

- **Step 2.2: Find $EU(a|e)\forall a \in A$**

$$VPI(E'|E = e) = MEU(E = e, E') - MEU(E = e)$$

$$EU(a|e) = \sum_{s \in S} P(s|a, e) \cdot U(s, a)$$

$$EU(B|T = t) = \sum_{q \in Q} P(Q = q|T = t) \cdot U(Q = q, B = b)$$

Case 1: T=1

$$EU(B = 0|T = 1) = P(Q = 0|T = 1) \cdot U(Q = 0, B = 0) + P(Q = 1|T = 1) \cdot U(Q = 1, B = 0) = 0.16 * 60 + 0.84 * 20 = 26.4$$

$$EU(B = 1|T = 1) = P(Q = 0|T = 1) \cdot U(Q = 0, B = 1) + P(Q = 1|T = 1) \cdot U(Q = 1, B = 1) = 0.16 * 40 + 0.84 * 100 = 90.4$$

$$MEU(T = 1) = 90.4 \;\&\; a^* = \{B = 1\}$$

Q	T	P(T|Q)	P(Q|T)
0	0	0.8	**0.64**
0	1	0.2	**0.16**
1	0	0.3	**0.36**
1	1	0.7	**0.84**

Q	B	U(Q,B)
0	0	**60**
0	1	**40**
1	0	**20**
1	1	**100**

Step 2: Compute $MEU(E = e, E')\forall e' \in E'$

- **Step 2.3: Find $\boldsymbol{MEU(E = e, E')}$** $VPI(E'|E = e) = MEU(E = e, E') - MEU(E = e)$
  - The weighted sum of $MEU(E = e, E' = e')$
  - Step 2.2: $MEU(T = 0) = 61.6$ and $MEU(T = 1) = 90.4$
  - $MEU(E = e, E') = \sum_{e'} P(E = e, E' = e') \cdot MEU(E = e, E' = e')$
    $= P(T = 0)MEU(T = 0) + P(T = 1)MEU(T = 1)$
    $= 0.5 * 61.6 + 0.5 * 90.4 = \mathbf{76}$

Step 3: Compute $VPI(E'|E = e)$

$$VPI(E'|E = e) = MEU(E = e, E') - MEU(E = e)$$

- Plug-in and Chug-in
  - Step 1: $MEU(E = e) = 76$
  - Step 2: $MEU(E = e, E') = 76$

$$VPI(E'|E = e) = MEU(E = e, E') - MEU(E = e)$$
$$= 76 - 76 = \mathbf{0}$$